Hybrid Control for Robust and Global Obstacle Avoidance via the Eikonal Equation

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***Abstract*— We consider the reach-avoid problem of steering a point-mass vehicle to a target set while avoiding obstacles, and present two control strategies: i) using a distance-to-target like function that is obtained by solving a Hamilton-Jacobi PDE, a subgradient descent controller ensures safe and almost-global finite-time stabilization of the target set; ii) this controller is extended using methods from hybrid control theory to achieve safe, robust, and global finite-time stabilization of the target set. Numerical simulations illustrate these results.**

I. Introduction

Obstacle avoidance, or *reach–avoid*, problems concern steering a dynamical system toward a desired target set while avoiding unsafe regions or obstacles in the state space. A widely used approach to solving the above problems is based on constructing a potential field via a Hamilton–Jacobi partial differential equation, most notably the *Eikonal equation*, whose solution encodes distance-to-target information while accounting for obstacles [1].

Eikonal-equation-based methods have shown considerable empirical success in motion planning and navigation problems [2], and their effectiveness has been demonstrated in numerous simulation-based studies [3]. Despite their widespread use, however, a rigorous theoretical analysis establishing whether such methods provably satisfy the fundamental objectives of obstacle avoidance remains largely absent from the literature. In the unconstrained setting, i.e., in the absence of obstacles, [4] leverages the Eikonal equation to design feedback controllers for point-mass dynamics and proves finite-time stability of the target set for the resulting closed-loop system. To the best of the authors' knowledge, a similar analysis in the presence of state constraints induced by obstacles remains undeveloped.

To this end, the objective of this paper is twofold. Firstly, we aim to bridge the above gap by providing a theoretical foundation for Eikonal-equation-based obstacle avoidance. We consider a broad class of static obstacles in the state space, without requiring convexity of the obstacles. By solving the Eikonal equation that requires information about the obstacles as its boundary condition, we obtain a distance-to-target function, leading to a subgradient descent controller that ensures almost global finite-time stabilization while avoiding the obstacles.

Secondly, noting the existence of the topological obstruction that renders the obstacle-free space noncontractible, a continuous feedback (such as the subgradient descent controller above) fails to ensure global convergence of solutions to the target set [5, Cor. 5.9.14]. In this setting, a globally stabilizing discontinuous feedback is necessarily nonrobust to arbitrarily small adversarial perturbations [6]. Therefore, we design a hybrid state-feedback controller with hysteretic switching to ensure robust and global finite-time stabilization of the target set while also ensuring obstacle avoidance.

Hybrid systems theory [7] has previously been used for obstacle avoidance problems [8], [9] to ensure safe, robust, and global *asymptotic* stabilization of the target set. The present paper improves these results by guaranteeing finite-time stabilization instead. The trade-off, however, is that our approach requires global knowledge of the obstacles to formulate boundary conditions for the Eikonal equation. In contrast, [8] only requires their local knowledge. Due to space constraints, proofs will be presented elsewhere.

II. Preliminaries

Let $\mathbb{N} = \{0, 1, 2 \ldots\}$, $\mathbb{R} \coloneqq (-\infty, \infty)$, $\mathbb{R}_{\geq 0} \coloneqq [0, \infty)$, and $\mathbb{B} \coloneqq \{v \in \mathbb{R}^n : |v| \leq 1\}$. Given a set $S \subset \mathbb{R}^n$, we denote its interior, boundary, and closed convex hull, respectively, by $\operatorname{int} S$, ∂S, and $\overline{\operatorname{co}} S$. The distance from $x \in \mathbb{R}^n$ to a closed set S is denoted by $|x|_S \coloneqq \inf_{y \in S} |x - y|$. The tangent cone to S at $x \in S$ is denoted by $\mathrm{T}_S(x) \coloneqq \limsup_{\tau \searrow 0} \frac{S-x}{\tau}$. Given a locally Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$, its (Clarke) generalized gradient is defined as $\partial f(x) \coloneqq \operatorname{co}\{\lim_{k\to\infty} \nabla f(x_k) : \exists (x_k \to x) \text{ s.t. } f \text{ is differentiable at } x_k\}$. A function $f : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ is positive definite with respect to $\mathcal{A} \subset \mathbb{R}^n$, denoted by $f \in \mathcal{PD}(\mathcal{A})$, if $f(x) = 0$ for each $x \in \mathcal{A}$, and $f(x) > 0$ otherwise. A function $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a class-$\mathcal{KL}$ function, written as $\beta \in \mathcal{KL}$, if it is nondecreasing in its first argument, nonincreasing in its second argument, $\lim_{r \searrow 0} \beta(r, s) = 0$ for each $s \geq 0$, and $\lim_{s \to \infty} \beta(r, s) = 0$ for each $r \geq 0$. A C^1-function $f : \mathbb{R}^n \supset X \to \mathbb{R}$ is denoted by $f \in C^1(X)$. A set-valued map $F : X \rightrightarrows Y$ assigns a set $F(x) \subset Y$ to each $x \in X$. Its domain is denoted by $\operatorname{dom} F \coloneqq \{x \in X : F(x) \neq \varnothing\}$.

We say that a set $H \subset \mathbb{R}^n$ is a plane if there exist $v \in \mathbb{R}^n \setminus \{0\}$ and $b \in \mathbb{R}$ such that $H = \{x \in \mathbb{R}^n : v^\top x + b = 0\}$. A normal vector to H in $\mathbb{R}^n$ is either ν or $-\nu$, where $\nu \coloneqq v/|v|$.

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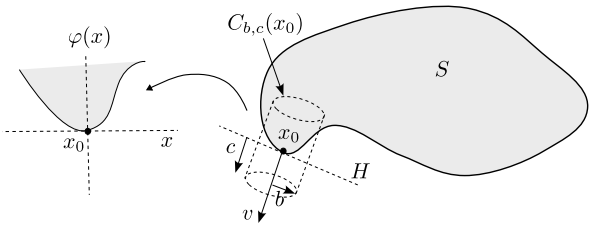


Fig. 1: A C^1-domain at x_0.

Definition 1 (C^1-domain)**.** A nonempty, open set $S \subset \mathbb{R}^n$ is said to be a C^1-*domain at* $x_0 \in \partial S$ if there exist constants $b, c > 0$, a plane $H \subset \mathbb{R}^n$ with $x_0 \in H$, a normal v to H, and an open cylinder $C_{b,c}(x_0) \coloneqq \{x + rv : x \in H, |x - x_0| < b, |r| < c\}$ such that $C_{b,c}(x_0) \cap S = C_{b,c}(x_0) \cap \{x + rv : x \in H, r > \varphi(x)\}$ and $C_{b,c}(x_0) \cap \partial S = C_{b,c}(x_0) \cap \{x + rv : x \in H, r = \varphi(x)\}$ for some C^1-function $\varphi : H \to \mathbb{R}$ such that $\varphi(x_0) = 0$ and $|\varphi(x)| < c$ for each $x \in x_0 + b\mathbb{B}$. The set S is a C^1-*domain* if it is a C^1-domain at each $x_0 \in \partial S$. Figure 1 illustrates the setup.

A. Hybrid Dynamical Systems

Following [7], a hybrid system $\mathcal{H}$ has solutions that may evolve continuously according to a differential inclusion defined by a set-valued map $F : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ when the state belongs to the *flow set* $C \subset \mathbb{R}^n$, and, when the state belongs to the *jump set* $D \subset \mathbb{R}^n$, solutions may jump according to a reset law defined by a set-valued map $G : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$. A hybrid system $\mathcal{H} = (C, F, D, G)$ is given as

$$\mathcal{H} : \begin{cases} \dot{x} & \in & F(x) & x \in C, \\ x^+ & \in & G(x) & x \in D, \end{cases} \tag{1}$$

where $x \in \mathbb{R}^n$ is the state. Solutions to $\mathcal{H}$ are parametrized by $(t, j) \in \mathbb{R}_{\geq 0} \times \mathbb{N}$, where $t \in \mathbb{R}_{\geq 0}$ denotes the time of continuous evolution of the solution and $j \in \mathbb{N}$ denotes the number of jumps occurred. Each solution to $\mathcal{H}$ is defined on a hybrid time domain which has the form $([0, t_1] \times \{0\}) \cup ([t_1, t_2] \times \{1\}) \cup \cdots$ for a nondecreasing sequence $0 \leq t_1 \leq t_2 \leq \ldots$, where the above series may be finite or infinite. If it is finite, the last interval of the hybrid time domain may have the form $[t_j, t_{j+1}]$, $[t_j, t_{j+1})$, or $[t_j, \infty)$.

A solution to $\mathcal{H}$ is a function $\phi : \operatorname{dom} \phi \to \mathbb{R}^n$ such that $\operatorname{dom} \phi$ is a hybrid time domain, for each $j \in \mathbb{N}$, the function $t \mapsto \phi(t, j)$ is locally absolutely continuous on the interval $I^j \coloneqq \{t : (t, j) \in \operatorname{dom} \phi\}$, $\phi(0, 0) \in \overline{C} \cup D$, and (i) for each $j \in \mathbb{N}$ such that $\operatorname{int} I^j$, $\phi(t, j) \in C$ for each $t \in \operatorname{int} I^j$ and $\frac{d\phi}{dt}(t, j) \in F(\phi(t, j))$ for almost all $t \in I^j$; and for each $(t, j) \in \operatorname{dom} \phi$ such that $(t, j + 1) \in \operatorname{dom} \phi$, we have $\phi(t, j) \in D$ and $\phi(t, j + 1) \in G(\phi(t, j))$.

A solution to $\mathcal{H}$ is *complete* if its domain is unbounded, and *maximal* if the solution cannot be extended to a function on a strictly larger domain that is also a solution to $\mathcal{H}$. We denote the set of maximal solutions to $\mathcal{H}$, and those from a nonempty set K, by $\mathcal{S}_{\mathcal{H}}$ and $\mathcal{S}_{\mathcal{H}}(K)$, respectively.

Next, we recall the following notion of finite-time stability of a nonempty, compact set $\mathcal{A} \subset \mathbb{R}^n$ for $\mathcal{H}$ from [10], and extend it to define global and robust (to small perturbations) versions of the notion on some closed set $\mathcal{X} \subset \mathbb{R}^n$.

Definition 2 (Finite-time stability)**.** Consider a hybrid system $\mathcal{H}$ on $\mathbb{R}^n$, a nonempty closed set $\mathcal{X} \subset \mathbb{R}^n$, a nonempty compact set $\mathcal{A} \subset \mathcal{X}$, and a functional $\mathcal{T} : \mathcal{S}_{\mathcal{H}}(\mathcal{X}) \to [0, \infty)$. The set $\mathcal{A}$ is said to be

1) *stable* for $\mathcal{H}$ if, for each $\varepsilon > 0$, there exists $\delta > 0$ such that, for each maximal solution ϕ to $\mathcal{H}$ with $|\phi(0, 0)|_{\mathcal{A}} \leq \delta$, we have that $|\phi(t, j)|_{\mathcal{A}} \leq \varepsilon$ for each $(t, j) \in \operatorname{dom} \phi$;
2) *almost globally finite-time attractive* on $\mathcal{X}$ for $\mathcal{H}$ if there exists a nonempty set $S \subset \mathcal{X}$ having Lebesgue measure zero in $\mathbb{R}^n$ such that, for each $\phi \in \mathcal{S}_{\mathcal{H}}(\mathcal{X} \setminus S)$ with $\sup_{(t,j) \in \operatorname{dom} \phi} t + j \geq \mathcal{T}(\phi)$, we have $\lim_{t+j \nearrow \mathcal{T}(\phi), (t,j) \in \operatorname{dom} \phi} |\phi(t, j)|_{\mathcal{A}} = 0$;
3) *globally finite-time attractive* on $\mathcal{X}$ if item 2 above holds with $S = \varnothing$;
4) *almost globally* (resp., *globally*) *finite-time stable* (FTS) on $\mathcal{X}$ for $\mathcal{H}$ if it is stable and almost globally (resp., globally) finite-time attractive on $\mathcal{X}$ for $\mathcal{H}$.
5) *robustly*[1] *and globally FTS* on $\mathcal{X}$ for $\mathcal{H}$ if
   a) $\mathcal{A}$ is globally finite-time stable on $\mathcal{X}$ for $\mathcal{H}$;
   b) there exists an open set $\mathcal{U} \subset \mathbb{R}^n$ containing $\mathcal{X}$ such that, for each continuous function $\rho : \mathbb{R}^n \to \mathbb{R}$ that is positive on $\mathcal{U} \setminus \mathcal{A}$ and each function $\beta \in \mathcal{KL}$ for which each $\phi \in \mathcal{S}_{\mathcal{H}}(\mathcal{U})$ satisfies, for all $(t, j) \in \operatorname{dom} \phi$,
   $$|\phi(t, j)|_{\mathcal{A}} \leq \beta(|\phi(0, 0)|_{\mathcal{A}}, t + j), \tag{2}$$
   the following holds: for each compact set $K \subset \mathcal{U}$ and each $\epsilon > 0$, there exists $\delta \in (0, 1)$ such that each solution $\phi_{\delta\rho}$ to the perturbed hybrid system $\mathcal{H}_{\delta\rho} = (C_{\delta\rho}, F_{\delta\rho}, D_{\delta\rho}, G_{\delta\rho})$ from K satisfies $|\phi_{\delta\rho}(t, j)|_{\mathcal{A}} \leq \beta(|\phi_{\delta\rho}(0, 0)|_{\mathcal{A}}, t + j) + \epsilon$ for each $(t, j) \in \operatorname{dom} \phi_{\delta\rho}$, where $C_{\delta\rho} \coloneqq \{x \in \mathbb{R}^n : (x + \delta\rho(x)\mathbb{B}) \cap C \neq \varnothing\}$, $D_{\delta\rho} \coloneqq \{x \in \mathbb{R}^n : (x + \delta\rho(x)\mathbb{B}) \cap D \neq \varnothing\}$, and for each $x \in \mathbb{R}^n$, $F_{\delta\rho}(x) \coloneqq \overline{\operatorname{co}} F((x + \delta\rho(x)\mathbb{B}) \cap C) + \delta\rho(x)\mathbb{B}$ and $G_{\delta\rho}(x) \coloneqq \{v \in \mathbb{R}^n : v \in g + \delta\rho(g)\mathbb{B}, g \in G((x + \delta\rho(x)\mathbb{B}) \cap D)\}$.

We use the following result in the forthcoming Theorem 2.

Lemma 1. *Suppose that $\mathcal{H}$ satisfies hybrid basic conditions.*[2] *Given a nonempty closed set $\mathcal{X} \subset \mathbb{R}^n$ and a nonempty compact set $\mathcal{A} \subset \operatorname{int} \mathcal{X}$, suppose that $\mathcal{A}$ is globally finite-time stable on $\mathcal{X}$ for $\mathcal{H}$. Then, $\mathcal{A}$ is robustly and globally finite-time stable on $\mathcal{X}$ for $\mathcal{H}$.*

Finally, we call a set $K \subset \mathbb{R}^n$ *forward pre-invariant* for $\mathcal{H}$ if each $\phi \in \mathcal{S}_{\mathcal{H}}(K)$ satisfies $\operatorname{rge} \phi \subset K$. The set K is *forward invariant* if, additionally, each $\phi \in \mathcal{S}_{\mathcal{H}}(K)$ is complete.

III. Problem Setup and Outline of Proposed Solution

Consider a vehicle with point-mass dynamics

$$\dot{x} = u, \tag{3}$$

[1]This robustness notion is in the semiglobal practical sense [7, Def. 7.18].

[2]Hybrid basic conditions (c.f. [7, Assumption 6.5]) are mild assumptions on the data of the hybrid system $\mathcal{H}$ that ensure well-posedness of $\mathcal{H}$. In particular, C, D are closed, F is outer semicontinuous and locally bounded relative to C, $F(x)$ is convex for each $x \in C$, $C \subset \operatorname{dom} F$, G is outer semicontinuous and locally bounded relative to D, and $D \subset \operatorname{dom} G$.

where $x \in \mathbb{R}^n$ denotes the position of the vehicle and $u \in \mathbb{R}^n$ denotes the control input. The objective is to design a feedback law assigning u that *robustly and globally* steers the vehicle position to a target set $\mathcal{A} \subset \mathbb{R}^n$ in finite time while avoiding the set of obstacles $\mathcal{O} \subset \mathbb{R}^n$ in the space. For brevity, we define the *safe set* $X_s \coloneqq \overline{\mathbb{R}^n \setminus \mathcal{O}}$. Formally, the problem that we solve is as follows.

Problem 1. Design a feedback controller $\kappa : \mathbb{R}^n \to \mathbb{R}^n$ for (3) that assigns u so that, for the resulting closed-loop system $\dot{x} = \kappa(x)$,

(r-FTS) $\mathcal{A}$ is robustly and globally finite-time stable on X_s;
(FI) the set X_s is forward invariant.

Recall from Definition 2 that the use of the term *global* above does not imply that all maximal solutions starting from any point in $\mathbb{R}^n$ arrive at $\mathcal{A}$ in finite time – instead, this property is only required to hold for solutions starting from X_s. A key challenge to global stabilization on X_s in the presence of obstacles is that the resulting safe space X_s becomes noncontractible.[3] This fact precludes the existence of a continuous static state-feedback control law for (3) that steers the closed-loop solutions to $\mathcal{A}$ – such a control law must be discontinuous. On the other hand, with a discontinuous control law ensuring global stabilization on X_s, the resulting stability guarantees can be destroyed by arbitrarily small adversarial perturbations [6].

To overcome the above challenges and solve Problem 1, we borrow ideas from hybrid control theory and the realm of optics. We propose to use the Eikonal equation, which is a PDE that describes the propagation of wavefronts of light that emanate from a light source in a medium with varying refractive index. We introduce the Eikonal equation in the forthcoming Section V. The detailed steps of our approach are as follows:

Step 1 We pose a particular calculus of variations problem that accounts for the obstacles and the admissible state space. The value function of this problem acts as a potential field that a subgradient descent controller uses to achieve finite-time stabilization; see Step 3.
Step 2 We show that the value function is a solution to the Eikonal equation. Therefore, it can be easily computed through numerical methods; see Sec. VIII.
Step 3 Next, we show that subgradient descent using the above value function ensures almost global finite-time stability of $\mathcal{A}$ on X_s for the closed-loop system, and satisfies (FI) in Problem 1. As this controller precludes global convergence, we call it the *local controller*.
Step 4 Finally, (r-FTS) in Problem 1 is achieved by designing a hybrid control law that switches between subgradient descent of multiple, appropriately designed solutions to the Eikonal equation.

To impose sufficient regularity on the types of admissible obstacles, we use the notion of a C^1-domain from Definition 1 and assume the following.

Assumption 1. The sets $\mathcal{A}, \mathcal{O} \subset \mathbb{R}^n$ satisfy the following:

(i) for each $x \in \mathcal{O}$ and each $\varepsilon > 0$, the set $(x + \varepsilon\mathbb{B}) \cap \mathcal{O}$ has positive Lebesgue measure in $\mathbb{R}^n$;
(ii) the safe set $X_s = \overline{\mathbb{R}^n \setminus \mathcal{O}}$ is nonempty, connected, and $\operatorname{int} X_s$ is a C^1-domain;
(iii) $\mathcal{A}$ is nonempty, compact, and $\mathcal{A} \subset \operatorname{int} X_s$.

We elaborate on the importance of item (i) in the forthcoming Remark 2.

Remark 1. The assumption in item (ii) requiring X_s to be a C^1-domain is quite general. For example, the set X_s would satisfy this assumption if the set of obstacles $\mathcal{O}$ has smooth boundaries. We do not impose strong assumptions like convexity of the obstacles. Situations when X_s does not satisfy item (ii) are when $\mathcal{O}$ partitions $\mathbb{R}^n$ so that X_s is disconnected and when $\mathcal{O}$ is a polytope. However, in the latter case, one can consider a smooth (over) approximation of the polytope so that the resulting X_s is a C^1-domain.

IV. Calculus of Variations and the Value Function

In this section, we pose the calculus of variations problem mentioned in Step 1. Specifically, we consider the minimization of the length of Lipschitz functions (curves) connecting any point $x \in X_s$ to any point y in the target set $\mathcal{A}$ while avoiding obstacles. To this end, consider the set of Lipschitz functions with speed no greater than one that connect each $x \in X_s$ to $\mathcal{A}$ in finite time $T \in [0, \tau_x]$, $\tau_x \geq 0$, denoted by

$$\mathcal{C}_{\leq \tau_x}(x) \coloneqq \left\{ \begin{array}{c} \gamma \in \operatorname{Lip}(\mathbb{R}_{\geq 0}, X_s) : \gamma(0) = x, \exists T \in [0, \tau_x] \\ \text{s.t. } \gamma(T) \in \mathcal{A}, \gamma(t) = \gamma(T) \ \forall t \geq T, \\ |\dot{\gamma}(t)| \leq 1 \text{ for almost all } t \geq 0 \end{array} \right\}$$

for each $x \in X_s$, where $\operatorname{Lip}(\mathbb{R}_{\geq 0}, X_s)$ denotes the set of Lipschitz functions $\gamma : \mathbb{R}_{\geq 0} \to X_s$. The subscript "$x$" in "$\tau_x$" highlights the fact that the value of τ_x depends on $x \in X_s$.

Lemma 2. *Suppose that Assumption 1 holds. Then, there exists a continuous function $\tau : X_s \to \mathbb{R}_{\geq 0}$ such that $\mathcal{C}_{\leq \tau_x}(x) \neq 0$ for each $x \in X_s$, where $\tau_x \coloneqq \tau(x) \geq 0$.*

Given $x \in X_s$ and $\gamma \in \mathcal{C}_{\leq \tau_x}(x)$ with τ_x from Lemma 2, we define the weighted length of γ by

$$J(\gamma) \coloneqq \int_0^\infty \frac{|\dot{\gamma}(t)|}{\mathfrak{n}(\gamma(t))} dt, \tag{4}$$

where the function $\mathfrak{n} : X_s \to \mathbb{R}_{>0}$ achieves normalization. We impose the following mild assumptions on $\mathfrak{n}$.

Assumption 2. The function $\mathfrak{n} : X_s \to \mathbb{R}_{>0}$ is Lebesgue measurable and there exist $0 < \underline{c} < \overline{c}$ such that $\operatorname{rge} \mathfrak{n} \subset [\underline{c}, \overline{c}]$.

For each $x \in X_s$ and τ_x from Lemma 2, define

$$V(x) \coloneqq \inf_{\gamma \in \mathcal{C}_{\leq \tau_x}(x)} J(\gamma) \qquad \forall x \in X_s. \tag{5}$$

We establish key properties of the value function V.

Proposition 1. *Suppose that Assumptions 1 and 2 hold. Then,*

[3] A set is noncontractible if it cannot be continuously shrunk down to a point in the set.

(i) $V \in \mathcal{PD}(\mathcal{A})$ *and* V *is locally Lipschitz on* X_s*;*

(ii) for each $x \in X_s$ *with* τ_x *according to Lemma 2, there exists* $\gamma^\star \in \mathcal{C}_{\leq \tau_x}(x)$ *such that* $V(x) = J(\gamma^\star)$ *and, for some* $T^\star \in [0, \tau_x]$, $|\dot{\gamma}^\star(t)| = 1$ *for almost all* $t \in [0, T^\star]$ *and* $\gamma^\star(t) = \gamma^\star(T^\star) \in \mathcal{A}$ *for each* $t \geq T^\star$.

Remark 2. Item (i) in Assumption 1 is crucial to prevent the shortest path γ from *passing through* a "thin" obstacle; i.e., one having Lebesgue measure zero. For example, in $\mathbb{R}^2$, let $\mathfrak{n}(x) \coloneqq 1$ for each $x \in \mathbb{R}^2$, $\mathcal{O} \coloneqq [-1, 1] \times \{1\}$, and $\mathcal{A} \coloneqq \{0\} \in \mathbb{R}^2$. Since $X_s = \overline{\mathbb{R}^2 \setminus \mathcal{O}}$, it follows that $X_s = \mathbb{R}^2$. Then, for $x = (0, 2) \in \mathbb{R}^2$, the shortest path in X_s connecting x to $\mathcal{A}$ is $\gamma(t) = (1 - t)x$ for each $t \in [0, 1]$, which passes through the obstacle. Item (i) in Assumption 1 prevents this from happening by rendering such an obstacle inadmissible.

Example 1. We illustrate Proposition 1 with a simple example where $\mathcal{O} \coloneqq \varnothing$ and $\mathcal{A} \coloneqq \{0\} \subset \mathbb{R}^n$. Then, $X_s \coloneqq \mathbb{R}^n$ and, with $\mathfrak{n}(x) \coloneqq 1$ for each $x \in X_s$, we have that $V(x) = |x|$ for each $x \in X_s$, thus verifying item (i) of Proposition 1. Next, for each $x \in X_s$, letting $\tau_x \coloneqq |x|$ and defining $\gamma^\star \in \mathcal{C}_{\leq \tau_x}(x)$ as $\gamma^\star(t) \coloneqq \left(1 - \frac{t}{|x|}\right)x$ for all $t \in [0, \tau_x]$ and $\gamma^\star(t) \coloneqq 0$ for all $t > \tau_x$, verifies item (ii). ■

V. Value Function and the Eikonal Equation

In this section, we follow Step 2 and establish a way to compute the value function V in (5). In particular, we show in forthcoming Proposition 2 that V is a solution, defined in an appropriate sense, to the so-called Eikonal equation, given by the PDE in (6) below. Then, numerical methods can be used to solve it and obtain V.

With the function $\mathfrak{n}$ satisfying Assumption 2, and sets X_s and $\mathcal{A}$ satisfying Assumption 1, consider the following PDE:

$$\begin{cases} |\nabla u(x)| = \dfrac{1}{\mathfrak{n}(x)} & x \in \operatorname{int}(X_s) \setminus \mathcal{A} \\ u(x) = 0 & x \in \partial\mathcal{A} \end{cases}. \tag{6}$$

This PDE is known as the Eikonal equation [4]. It is a Hamilton-Jacobi equation, whose solutions are defined in the *viscosity sense*. Before presenting the notion of viscosity solutions, we define $H : \mathbb{R}^n \times (\operatorname{int}(X_s) \setminus \mathcal{A}) \to \mathbb{R}$ as $H(p, x) \coloneqq |p| - 1/\mathfrak{n}(x)$ for each $(p, x) \in \mathbb{R}^n \times (\operatorname{int}(X_s) \setminus \mathcal{A})$. We begin by assuming additional regularity on $\mathfrak{n}$.

Assumption 3. The function $\mathfrak{n} : X_s \to \mathbb{R}_{>0}$ is continuous.

Now, using [11, Def. 2.5] and [12, Ch. III, (3.23)], we present the following notion of a viscosity solution to (6). For brevity, define $\Gamma \coloneqq (\operatorname{int}(X_s) \setminus \mathcal{A}) \cup \partial\mathcal{A}$.

Definition 3 (Viscosity solution)**.** An upper semicontinuous function $u : \Gamma \to \mathbb{R}$ is a *viscosity subsolution* to (6) if, for each $\varphi \in C^1(\operatorname{int}(X_s) \setminus \mathcal{A})$ such that $\varphi(x) \geq u(x)$ for each $x \in \operatorname{int}(X_s) \setminus \mathcal{A}$, $H(\nabla\varphi(x_0), x_0) \leq 0$ for each $x_0 \in \operatorname{int}(X_s) \setminus \mathcal{A}$ satisfying $\varphi(x_0) = u(x_0)$, and $u(x) \geq 0$ for each $x \in \partial\mathcal{A}$. A lower semicontinuous function $u : \Gamma \to \mathbb{R}$ is a *viscosity supersolution* to (6) if, for each $\psi \in C^1(\operatorname{int}(X_s) \setminus \mathcal{A})$ such that $\psi(x) \leq u(x)$ for each $x \in \operatorname{int}(X_s) \setminus \mathcal{A}$, $H(\nabla\psi(x_0), x_0) \geq 0$ for each $x_0 \in \operatorname{int}(X_s) \setminus \mathcal{A}$ satisfying $\psi(x_0) = u(x_0)$, and $u(x) \leq 0$ for each $x \in \partial\mathcal{A}$. A continuous function $u : \Gamma \to \mathbb{R}$ is a *viscosity solution* to (6) if it is a viscosity subsolution and supersolution to (6).

Now, we show that V is a viscosity solution to (6).

Proposition 2. *Suppose that Assumptions 1, 2, and 3 hold. Then, the restriction of the value function* V *to* Γ*, denoted by* $V|_\Gamma : \Gamma \to \mathbb{R}$*, is a viscosity solution to* (6)*. Furthermore, if* $\mathfrak{n}$ *is Lipschitz, then* $V|_\Gamma$ *is the unique viscosity solution to* (6)*.*

Remark 3. As $V|_\Gamma$ is a viscosity solution to the Eikonal equation, several numerical methods to solve (6) can be used to obtain a function $U = V|_\Gamma$ with $\operatorname{dom} U = \Gamma$; see [13], [14]. Then, $\operatorname{dom} U$ can be artificially extended to X_s by letting $U|_{\mathcal{A}} \equiv 0$, so that U and V agree on $\operatorname{int} X_s$, and letting $U(x) \coloneqq \lim_{y \to x, y \in \operatorname{int} X_s} U(y)$ for each $x \in \partial X_s$, where the limit exists as $U = V|_\Gamma$ is locally Lipschitz on $\operatorname{int} X_s$.

VI. Local Controller Design and Lyapunov Analysis

In this section, we follow Step 3 and use the value function V to design a control law that ensures almost global finite-time stabilization of $\mathcal{A}$ on X_s and forward invariance of X_s for the resulting closed-loop system. For the latter property, in addition to designing a control law on X_s, one also needs to appropriately design a control law in a neighborhood outside X_s so that solutions starting outside X_s from this neighborhood are not pushed "away" from X_s; see [15, Ch. 4.6, Thm. 2] and the preceding discussion therein. We let this neighborhood be the open set $\mathcal{U}$ containing X_s so that each $x \in \mathcal{U} \setminus X_s$ has a unique projection onto ∂X_s; such a $\mathcal{U}$ exists as $\operatorname{int} X_s$ is a C^1-domain.

For each $x \in X_s$, we assign u via a measurable function $\kappa : X_s \to \mathbb{R}^n$ which is a selection from $-\partial V$, namely, $\kappa(x) \in -\partial V(x)$ for each $x \in X_s$. For each $x \in \mathcal{U} \setminus X_s$, we assign u as the negative gradient of $B : \mathcal{U} \to \mathbb{R}$ that, for each $x \in \mathcal{U}$, is defined as $B(x) \coloneqq -|x|_{\partial X_s}$ if $x \in X_s$ and $B(x) \coloneqq |x|_{\partial X_s}$ if $x \in \mathcal{U} \setminus X_s$. As B is differentiable at each $x \in \mathcal{U} \setminus X_s$, we assign

$$u = \kappa'(x) \coloneqq \begin{cases} \kappa(x) & \text{if } x \in X_s \\ -\nabla B(x) & \text{if } x \in \mathcal{U} \setminus X_s \end{cases} \qquad \forall x \in \mathcal{U}, \tag{7}$$

where recall that κ is a measurable selection from $-\partial V$, which exists by [16, Prop. 2.1.2(a)] and [16, Thm. 3.1.1].

Then, measurability of $\kappa' : \mathcal{U} \to \mathbb{R}^n$ above follows from [17, Ch.3, Prop.5]. However, measurability of κ' alone is not sufficient to ensure robustness of the resulting closed-loop system to small state perturbations. To achieve this property, we take the Krasovskii regularization [7, Ch. 4] of κ' and define the set-valued feedback $\widehat{\kappa}' : \overline{\mathcal{U}} \rightrightarrows \mathbb{R}^n$ by

$$\widehat{\kappa}'(x) \coloneqq \bigcap_{\delta > 0} \overline{\operatorname{co}}\, \kappa'((x + \delta\mathbb{B}) \cap \mathcal{U}) \qquad \forall x \in \overline{\mathcal{U}}. \tag{8}$$

The regularized feedback $\widehat{\kappa}'$ has the following properties.

Lemma 3. *Suppose that Assumptions 1 and 2 hold. Then,*

(i) $\widehat{\kappa}'$ *is outer semicontinuous and locally bounded relative to* $\overline{\mathcal{U}}$*, and, for each* $x \in \overline{\mathcal{U}}$*,* $\widehat{\kappa}'(x)$ *is convex;*

(ii) for each $x \in X_s$*,* $\widehat{\kappa}'(x) \subset \mathrm{T}_{X_s}(x)$*.*

Remark 4. Item (ii) of Lemma 3 ensures that, under the action of (8), the vehicle never collides "head-on" with the obstacles as its velocity is always tangential to the safe set X_s – these directions are characterized by the tangent cone to X_s. The vehicle, however, may "graze" the obstacle boundary. This phenomenon depends on the chosen $\mathfrak{n}$. In practice, we can artificially inflate the true obstacles to obtain the set $\mathcal{O}$ so that such "grazing" occurs at a safe distance from the true obstacles. Such inflation is illustrated in the forthcoming example in Section VIII via the smooth (over) approximation mentioned therein.

Using (8) with (3), consider the closed-loop system

$$\dot{x} \in \widehat{\kappa}'(x) \qquad x \in \overline{\mathcal{U}}. \tag{9}$$

To characterize the set of points in X_s from where solutions to (9) may not converge to $\mathcal{A}$ in finite time, let

$$S(V) \coloneqq \{x \in X_s : \operatorname{card}(\partial V(x)) \neq 1\} \setminus \mathcal{A}. \tag{10}$$

The set $S(V)$ has Lebesgue measure zero in $\mathbb{R}^n$. Next, we analyze stability properties of the set $\mathcal{A}$ for the closed-loop system (9). To do so, recall Definition 2 and note that (9) is a hybrid system with the jump set as the empty set.

Theorem 1. *Suppose that Assumptions 1 and 2 hold. Then,*

(i) the set X_s is forward invariant for (9)*;*

(ii) if the set $X_s \setminus S(V)$ is forward invariant for (9)*, then $\mathcal{A}$ is almost globally finite-time stable on X_s for* (9)*.*

Theorem 1 establishes the desired result by using the set-valued feedback law $\widehat{\kappa}'$. If using the single-valued, measurable κ' from (7), one cannot guarantee existence of nontrivial solutions, which results in the following weaker result.

Corollary 1. *Suppose that Assumptions 1 and 2 hold. Then, for $\dot{x} = \kappa'(x)$ $x \in \mathcal{U}$ with κ' from* (7)*, X_s is forward pre-invariant and, if $X_s \setminus S(V)$ is forward pre-invariant as well, then $\mathcal{A}$ is almost globally finite-time stable on X_s.*

VII. Hybrid Feedback Control for Global Obstacle Avoidance

Following the obstruction to global stabilization that was discussed in Section III, we follow Step 4 to design a hybrid feedback control law that solves Problem 1. The design of said hybrid control law leverages the local control design in (7) and uses ideas from synergistic hybrid control [18].

Let Q be a finite index set. Consider a family $\{\mathfrak{n}_q\}_{q\in Q}$ of functions $\mathfrak{n}_q : X_s \to \mathbb{R}_{>0}$ that satisfy Assumption 2. Due to (5), the above family of functions results in a family $\{V_q\}_{q\in Q}$ of value functions to the respective calculus-of-variations problems. Inspired by synergistic control [18], we assume the following.

Assumption 4 (Synergy gap)**.** There exist functions $\{\mathfrak{n}_q : X_s \to \mathbb{R}_{>0}\}_{q\in Q}$ satisfying Assumption 2 such that the respective value functions $\{V_q : X_s \to \mathbb{R}_{\geq 0}\}_{q\in Q}$ obtained from (5) satisfy $S(V_q) \neq \varnothing$ for each $q \in Q$ and $\mu \coloneqq \inf_{q\in Q, x\in S(V_q)}\{V_q(x) - \min_{p\in Q} V_p(x)\} > 0$, where, for each $q \in Q$, $S(V_q)$ is as defined in (10).

Remark 5. Assumption 4 ensures that, for each logic variable $q \in Q$ and each point $x \in S(V_q)$ from where the gradient descent using the function V_q may not steer solutions to $\mathcal{A}$, there exists another logic variable $p \in Q$ such that $x \notin S(V_p)$, and $V_p(x)$ is smaller than $V_q(x)$ by at least an amount μ.

Pick any $\delta \in (0, \mu)$. For each $q \in Q$, let κ_q be a measurable selection from $-\partial V_q$. Define $\kappa'_q : \mathcal{U} \to \mathbb{R}^n$ and $\widehat{\kappa}'_q : \overline{\mathcal{U}} \rightrightarrows \mathbb{R}^n$ as in (7) and (8), respectively, with κ therein replaced by κ_q. Following Theorem 1, for each $q \in Q$, the set-valued control law $\widehat{\kappa}'_q$ in (8) renders X_s forward invariant and $\mathcal{A}$ almost globally finite-time stable for the resulting closed-loop system. Then, designing a controller that appropriately toggles the value of q, leading to a different value of u as above, can result in global (and robust) finite-time stability of $\mathcal{A}$. To this end, we propose a hybrid controller $\mathcal{H}_K$ with state $q \in Q$, input $x \in \mathcal{U}$, and output ζ given by

$$\mathcal{H}_K : \begin{cases} \dot{q} &= F_K(x,q) & (x,q) \in C_K \\ q^+ &\in G_K(x,q) & (x,q) \in D_K \\ \zeta &= \widehat{\kappa}'_q(x) \end{cases}$$

where $C_K \coloneqq \{(x,q) \in X_s \times Q : V_q(x) - \min_{p\in Q} V_p(x) \leq \delta\}$, $D_K \coloneqq \overline{(X_s \times Q) \setminus C_K}$, $F_K(x,q) \coloneqq 0$ for all $(x,q) \in C_K$, and $G_K(x,q) \coloneqq \arg\min_{p\in Q} V_p(x)$ for all $(x,q) \in D_K$. The parameter δ denotes the *hysteresis gap*. The controller $\mathcal{H}_K$ switches q to $p \in Q$ if $V_p(x)$ is at least smaller than $V_q(x)$ by an amount δ; otherwise, $\mathcal{H}_K$ retains the current value of q.

The above hybrid controller, when applied to the continuous-time plant (3) so that the output ζ of $\mathcal{H}_K$ acts as the input u to (3), yields a hybrid closed-loop system $\mathcal{H} = (C_K, F, D_K, G)$ of the form (1) with state $z \coloneqq (x,q) \in \overline{\mathcal{U}} \times Q$, $F(z) \coloneqq (\widehat{\kappa}'_q(x), 0)$ for each $z \in C_K$, and $G(z) \coloneqq \{(x,p) : p \in G_K(x,q)\}$ for each $z \in D_K$.

Theorem 2. *Suppose that Assumptions 1, 2, and 4 hold. Then,*

(i) $X_s \times Q$ is forward invariant for $\mathcal{H}$;

(ii) $\mathcal{A} \times Q$ is robustly and globally FTS on $X_s \times Q$ for $\mathcal{H}$.

VIII. Numerical Details and Simulation Results

Due to the lack of a closed-form viscosity solution to the Eikonal equation (6) in the case of nontrivial obstacles $\mathcal{O}$ and the target set $\mathcal{A}$, we use [14] to numerically obtain a viscosity solution V to (6) that satisfies $V \in \mathcal{PD}(\mathcal{A})$.

A. Local Controller Implementation

Let $\mathcal{A} \coloneqq (1/2, 1/2) \in \mathbb{R}^2$ and $\mathcal{O} \subset \mathbb{R}^2$ be the union of the white, nonconvex polygon in Fig. 2 and the set $\mathbb{R}^2 \setminus (0,1)^2$. While $\mathcal{O}$ is a polytope, the following implementation uses a smooth (over) approximation of it which causes Assumption 1 to hold; see Remark 1. We define $\mathfrak{n} : X_s \to \mathbb{R}_{>0}$ as $\mathfrak{n}(x) \coloneqq 1/(1 + (1 + \varepsilon x_1)\exp(\frac{-|x|_{\partial X_s}}{\sigma}))$ for each $x \in X_s$, with $\sigma = 5$ and $\varepsilon = 10$, and note that Assumption 2 holds. Fig. 2 denotes the resulting unique viscosity solution to (6), and a sample solution to (9) that safely converges to $\mathcal{A}$. The above $\mathfrak{n}$ pushes the solution away from the obstacle boundary, which prevents the grazing phenomenon from Remark 4.

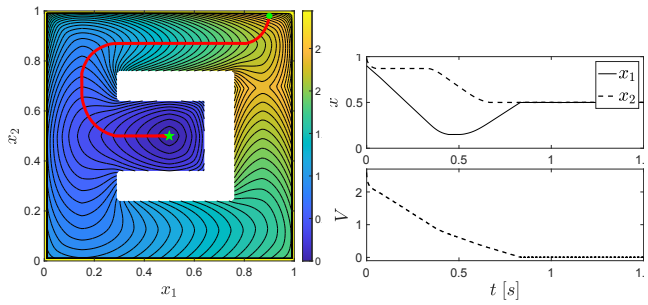


Fig. 2: *Left:* Contour plot of V on X_s, and a closed-loop solution x in red. The initial condition and the set $\mathcal{A}$ are denoted by a green circle and a green star, respectively. *Right:* Plot of V and x vs. time. Both coordinates of x converge, in finite time, to $\mathcal{A} = (1/2, 1/2)$. The function V decreases along x until it reaches $\mathcal{A}$ and remains at zero thereafter.

B. Hybrid Global Controller Implementation

Let $Q := \{-1, 1\}$ and, for each $q \in Q$, define $\mathfrak{n}_q : X_s \to \mathbb{R}_{>0}$ as $\mathfrak{n}_q(x) := 1/(1 + (1 + \varepsilon f(x,q)) \exp(-|x|_{\partial X_s}/\sigma))$ for each $x \in X_s$, where, using the fact that $X_s \subset [0,1]^2$, we define $f(x,q) := (1+q)x_1/2 + (1-q)(1-x_1)/2$ for each $(x,q) \in X_s \times Q$. Computing the viscosity solutions V_q, $q \in Q$, we verify via simulations that Assumption 4 holds with $\mu \approx 0.6$. Then, with $\delta = 0.01$, we simulate a solution to $\mathcal{H}$ in Fig. 3 that starts near $S(V_{-1})$, and observe that it converges to $\mathcal{A}$ in finite time, with a jump at $t = 0$.

IX. Conclusion

For the reach-avoid problem in the presence of a broad class of static obstacles, we employ the Eikonal equation to construct two controllers for a point-mass system in $\mathbb{R}^n$. Both controllers use the viscosity solution to define a gradient-descent feedback law. The first is a local controller that ensures almost global finite-time stability of the compact target set $\mathcal{A}$ while avoiding obstacles. The second is a hybrid controller that guarantees robust global finite-time stability of $\mathcal{A}$, thereby overcoming the topological obstruction encountered by the local controller. These results provide theoretical justification for a broad literature that employs the Eikonal equation for motion planning without formal guarantees.

Future work includes studying robustness of the proposed controller to a broader class of disturbances using input-to-state stability notions, as well as robustness to partial obstacle information that yields an imprecise solution to the Eikonal equation. Additional directions also include modeling second-order vehicle dynamics and accounting for moving obstacles, which may yield a time-varying PDE instead.

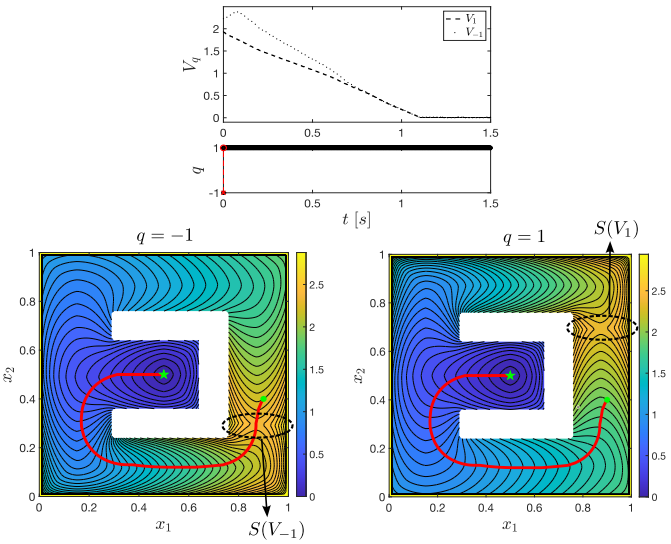


Fig. 3: *Top:* Plot of V_q and q vs. time. One jump at $t = 0$ is observed. Solution converges to $\mathcal{A}$ in finite time. *Bottom left:* Contour plot of V_{-1} on X_s. The solution is initialized close to $S(V_{-1})$, causing a jump at $(t,j) = (0,0)$. Then, $\operatorname{rge} x$ is not orthogonal to contours of V_{-1} as the hybrid controller uses the function V_1 instead. *Bottom right:* Contour plot of V_1 on X_s, with the same solution overlaid. Now, $\operatorname{rge} x$ is orthogonal to contours of V_1.

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